Layer Selection Matters: How Metric Choice Determines Hidden-State Probe Effectiveness for Reasoning-Model Safety Monitoring

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Abstract

Reasoning-trace language models can "self-jailbreak" during extended chain-of-thought generation, transitioning from safe reasoning to harmful content at a *safety commitment point*. We present the first systematic comparison of detection modalities for this phenomenon, evaluating hidden-state (HS) probes, text-based probes, sequence models, change-point detectors, and unsupervised methods across four models spanning two architectures (Llama, Qwen), two training methods (R1-distilled, STILL), and two scales (7–8B, 32B). Our primary methodological finding is that the composite layer-selection metric `bal_acc × crossing_rate` systematically biases toward shallow layers at 32B scale, entirely masking hidden-state probe advantages. This bias arises from the interaction between sequential probe scores and the crossing-rate statistic, and is specific to temporal monitoring pipelines where threshold-crossing metrics are used for layer selection. At correctly selected layers, HS probes significantly outperform text probes on all four models (12/12 Holm–Bonferroni tests pass, precision $+15$–24 pp). However, cross-model transfer fails entirely, six alternative detection methods fail, and no operating point simultaneously achieves $>70\%$ precision with $>60\%$ coverage on any model, establishing fundamental limitations of current probe-based safety monitoring. A boundary-condition analysis on AdvBench binary classification confirms that the layer selection bias does not manifest outside temporal monitoring settings, bounding the scope of our methodological contribution.

1 Introduction

Large reasoning models (DeepSeek-R1, QwQ) generate extended chains of thought (CoT) that can span hundreds of reasoning steps. Recent work has documented a "self-jailbreak" phenomenon in which these models' own reasoning processes erode safety guardrails during generation, culminating in a *safety commitment point* after which the model produces harmful content (**?**). Unlike adversarial-input attacks on instruction-tuned models, where hidden-state detectors achieve $>95\%$ defense rates (**?**), self-jailbreak originates *within* the model's reasoning process, and no prior work has systematically evaluated detection methods for this threat.

We address this gap with a comprehensive empirical study spanning four models (R1-8B, OpenThinker-7B, QwQ-32B, R1-32B), comparing hidden-state probes, text-based probes (Sentence-BERT embeddings), GRU sequence models, Bayesian change-point detection (BOCPD), CUSUM, Mahalanobis distance, and fine-grained taxonomy supervision. Our contributions are:

1. **Layer selection sensitivity** (methodological): The composite metric `bal_acc × crossing_rate` systematically selects shallow layers for 32B models due to noise-driven threshold crossings at early layers. This bias is specific to temporal monitoring pipelines using crossing-rate statistics; it does not manifest in binary classification settings (confirmed via AdvBench replication, §5.3). Correcting the layer selection reverses the conclusion from "HS advantage is scale-dependent" to "HS advantage is universal."
2. **HS probes outperform text probes at correctly selected layers**: 12/12 Holm–Bonferroni corrected tests pass across four models, with precision gains of $+15.4$ to $+23.6$ pp. Practical trade-offs are model-specific: 7–8B models show strong advantages with comparable timing; 32B models exhibit divergent lead-time and coverage profiles.
3. **Comprehensive negative results**: Six alternative methods fail—Mahalanobis distance, BOCPD, CUSUM, cross-model transfer, non-

linear probes (GRU/MLP), and fine-grained supervision—establishing that supervised temporal linear probing with principled layer selection is the most effective current approach.

2 Related Work

Hidden-state safety monitoring. TrajGuard (**?**) uses trajectory-level hidden-state features for adversarial-input defense on instruction-tuned models. SafeSwitch (**?**) and SIREN (**?**) similarly target input-level attacks. These methods operate at the token or utterance level and have not been evaluated on reasoning-model self-jailbreak, where the threat is the model's own extended reasoning.

Reasoning-model safety. Self-Jailbreaking (**?**) documents the phenomenon of safety erosion during CoT generation. HarmThoughts (**?**) characterizes harmful reasoning patterns. SafeThink (**?**) and SafeRemind (**?**) propose text-level CoT monitors but remain vulnerable to pathological reasoning traces. None of these works provide systematic modality comparisons or hidden-state probe evaluation.

Probing methodology. Linear probing of neural representations has been extensively studied in NLP (**?**). Layer selection is typically treated as a hyperparameter; we show it constitutes a significant researcher degree of freedom when composite metrics introduce systematic bias.

3 Method

3.1 Task: Safety Commitment Detection

Given a reasoning trace $\mathbf{x} = (x_1, \ldots, x_T)$ segmented into S reasoning steps, and a *commitment point* annotation s^* after which safety reasoning permanently ceases, the task is binary classification at each step s_t: predict whether $t \geq s^*$. We use commitment-point annotations from **?** across four reasoning models.

3.2 Probing Framework

Hidden-state probes. For each reasoning step s_t, we extract the hidden-state vector $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^d$ at layer ℓ (last token of the step). A logistic regression probe with balanced class weights maps $\mathbf{h}_t^{(\ell)}$ to a safety commitment prediction.

Text probes. We encode each reasoning step using Sentence-BERT (`all-MiniLM-L6-v2`) to obtain $\mathbf{e}_t \in \mathbb{R}^{384}$, and train an analogous logistic regression probe.

Temporal windowing. For window size W, the input at step t is the concatenation $[\mathbf{v}_{t-W+1}; \ldots; \mathbf{v}_t]$ where $\mathbf{v}$ is either $\mathbf{h}^{(\ell)}$ or $\mathbf{e}$. We evaluate $W \in \{1, 3, 5, 10, 15, 20, 25\}$.

3.3 Layer Selection Methodology

We evaluate five layer-selection metrics: (1) `bal_acc × crossing_rate`: the product of balanced accuracy and the rate at which probe scores cross the detection threshold; (2) pure `bal_acc`: threshold-free balanced accuracy; (3) `precision`: positive predictive value at $\tau = 0.5$; (4) `pre_cp_fpr`: inverse false-positive rate before the commitment point; (5) `composite`: additive combination of normalized precision and inverse FPR. We perform a full-layer sweep on all models and compare selected layers across metrics.

3.4 Detection Criterion

A commitment is detected at step t if at least K of the preceding W steps have probe scores exceeding threshold $\tau = 0.5$. We use $K = 1$ (single-crossing) as the default and report coverage (fraction of traces where detection fires), lead time (steps before true commitment point), and false-positive rate on safe traces.

3.5 Alternative Methods

We additionally evaluate: (a) **Mahalanobis distance** on safe-trace covariance (TrajGuard-style); (b) **BOCPD** and **CUSUM** on probe score sequences; (c) **GRU sequence models** on multi-layer hidden states; (d) **16-label fine-grained taxonomy** supervision; (e) **Cross-model transfer** via PCA-aligned probes.

4 Experiments

4.1 Models and Data

We evaluate four models in a 2×2 design: R1-8B (Llama-8B, R1-distilled, n_{test}=43), OpenThinker-7B (Qwen-7B, STILL-trained, n_{test}=47), QwQ-32B (Qwen-32B, native, n_{test}=19–20), and R1-32B (Qwen-32B, R1-distilled, n_{test}=20). Data splits are 60/20/20 with balanced class weights.

4.2 Layer Selection Sensitivity (C1)

Table 1 shows that the `bal_acc × crossing_rate` metric selects dramatically

Metric	R1-8B	OT-7B	QwQ	R1-32B
bal_acc × cr	L0 (0%)	L0 (0%)	L2 (3%)	L10 (16%)
pure bal_acc	L12 (39%)	L9 (32%)	L30 (47%)	L47 (73%)
precision	L16 (52%)	L16 (57%)	L60 (94%)	L63 (98%)
pre_cp_fpr	L16 (52%)	L16 (57%)	L60 (94%)	L63 (98%)
composite	L16 (52%)	L12 (43%)	L60 (94%)	L63 (98%)

Table 1: Layer selected by each metric per model. The `bal_acc × crossing_rate` metric selects shallow layers (0–16% depth for 32B models), while threshold-dependent metrics converge on deep layers (94–98%).

Model	Layer	W=1	W=3	W=5	W=10	Trend
R1-32B	L63	.826	.754	.741	.737	↓
QwQ-32B	L60	.740	.740	.742	.735	→
R1-8B	L14	.778	.790	.781	.793	↑
OT-7B	L16	.663	.688	.678	.668	↑↓

Table 2: Temporal window effect on HS probe BALACC at corrected layers. R1-32B declines with $W > 1$ due to dimensionality ($51200 \times W$ features, $n = 20$).

Model	W=1	W=3	W=15	W=25
R1-8B	.693/.721	.723/.698	.777/.512	.822/.488
OT-7B	—	.466/.532	—	—
R1-32B	.549/.800	.560/.800	.591/.700	.566/.700
QwQ-32B	.544/.789	.565/.737	.582/.684	.618/.579

Table 3: Precision/Coverage at key window sizes. No operating point achieves >70% precision with >60% coverage simultaneously.

Rk	Model	Metric	p	HB Thr.	Pass
1	R1-8B	precision	<.001	.0042	✓
2	OT-7B	bal_acc	<.001	.0045	✓
3	OT-7B	precision	<.001	.0050	✓
4	OT-7B	FPR	<.001	.0056	✓
5	QwQ	bal_acc	<.001	.0063	✓
6	QwQ	precision	<.001	.0071	✓
7	R1-32B	bal_acc	<.001	.0083	✓
8	R1-32B	precision	<.001	.0100	✓
9	QwQ	FPR	.004	.0125	✓
10	R1-8B	bal_acc	.005	.0167	✓
11	R1-32B	FPR	.007	.0250	✓
12	R1-8B	FPR	.044	.0500	✓

Table 4: Holm–Bonferroni step-down correction across 4 models × 3 metrics. All 12/12 tests pass at $\alpha = 0.05$. Precision advantage: $+15.4$ to $+23.6$ pp.

shallower layers than all other metrics, particularly for 32B models (L0–L10, 0–16% depth vs. L47–L63, 73–98% depth for threshold-dependent metrics).

Under leave-one-out cross-validation, the dominant layer is selected in 65–68% of folds across all metrics, confirming selection stability. The root cause is that shallow layers have more compressed representations, generating noise-driven threshold crossings that inflate the crossing-rate component.

4.3 Cross-Architecture Layer Sweep

Full layer sweeps on R1-32B (64 layers) and QwQ-32B (64 layers) confirm the bias. At biased layers, R1-32B achieves BALACC = 0.718 (L10), improving to 0.826 at the corrected L63. QwQ-32B achieves BALACC = 0.73 at biased L2, with 0.74 at corrected L60. For QwQ, the precision gain from correcting layer selection is $+11.5$ pp; for R1-32B, $+4.1$ pp.

4.4 Temporal Window Evaluation

Table 2 shows that temporal windowing effects are model-specific for HS probes: R1-8B improves with larger W, R1-32B degrades (dimensionality issue: $51200 \times W$ features for $n = 20$), and QwQ-32B is flat. Text probes universally benefit from temporal context across all four models.

4.5 Window Size Ablation

Table 3 characterizes the precision–coverage trade-off. Precision monotonically increases with W while coverage monotonically decreases. No model achieves simultaneous >70% precision and >60% coverage at any window size, establishing a fundamental practical limitation.

4.6 HS vs. Text: Bootstrap and Holm–Bonferroni

Using 10,000 bootstrap resamples, we test whether HS probes outperform text probes on balanced accuracy, precision, and pre-CP FPR across all four models. Table 4 shows the full Holm–Bonferroni step-down results: all 12/12 tests pass, with precision gains in the $+15.4$ to $+23.6$ pp range. Marginal tests (ranks 11–12) have wider confidence intervals, particularly for 32B models where $n \leq 20$.

HS vs. Text on CP Data. The text baseline achieves BALACC of 0.661–0.678 on R1-8B, 0.614 on OT-7B, 0.65 on QwQ-32B, and 0.646 on R1-32B. Text precision ranges from 0.329 to 0.487 across models. HS probes uniformly outperform text by $+15.4$ to $+23.6$ pp on precision, with all differences statistically significant after Holm–Bonferroni correction. Text requires zero model-

Method	BA	Prec	FPR	Cov	Lead
HS temp. W=3 L16	.688	.466	.100	.532	+21.6
Text temp. W=25	.614	.329	.161	.660	+45.5
GRU multi L13–19	.746	—	.274	.645	+33.6
HS–Text Δ (HB)	+10.7	+21.9	−17.1	—	—

Table 5: OpenThinker-7B pipeline summary ($n = 228$ traces, 47 test). HS–Text differences: all 3/3 HB pass. Selectivity = 18.3 pp (highest across models).

Model	Oracle	Best Probe	Method	Gap
R1-8B	+31.0	+4.5	GRU	26.5
OT-7B	∼+30	+33.6	GRU	∼0
QwQ-32B	+27	−1.2 to +1.9	Temp. LR	25–28
R1-32B	+28	+13.3	Temp. LR W=1	15

Table 7: Oracle vs. best probe lead time. Labels show gradual safety erosion (autocorrelation 0.82, entropy 1.38); hidden states show step-function transitions (autocorrelation 0.95, entropy 0.41).

Model	HS Lead	Text Lead	HS BA	HS > Pos?
R1-8B (W=15)	−0.1	+10.3	.802	Yes (100%)
OT-7B (W=3)	+21.6	+45.5	.688	No
QwQ-32B (W=3)	−1.2 to +1.9	+3.7 to +12.3	.740	No
R1-32B (W=1)	+11.7 to +13.3	—	.826	Yes (25%)

Table 6: Lead time (steps before commitment point). HS lead is model-dependent: R1-8B detects *at* commitment; QwQ detects later than text; R1-32B achieves positive lead but with 25% coverage.

Train →	R1-8B	OT-7B	R1-32B	QwQ
R1-8B	.817	—	.522	.595
OT-7B	—	.688	—	—
R1-32B	.583	—	.826	.673
QwQ-32B	—	—	.673	.740

Table 8: Cross-model transfer accuracy. Best transfer (0.673) is far below within-model performance. Cross-family (Llama↔Qwen) is near chance.

internal infrastructure, representing a meaningful practical baseline.

4.7 OpenThinker-7B Full Pipeline

OpenThinker-7B (Qwen2.5-7B, STILL-trained) provides the cleanest single-model evidence with the largest dataset ($n = 228$ traces). Table 5 shows 3/3 Holm–Bonferroni passes with the highest selectivity (18.3 pp) across all models. HS temporal probes saturate at $W = 3$; the GRU achieves higher balanced accuracy (0.746) but at substantially higher FPR (0.274 vs. 0.100).

4.8 Lead Time Analysis

Table 6 reveals heterogeneous timing profiles. R1-8B HS detects at the commitment point (lead ≈ 0), while text detects ∼10 steps earlier (inflated by higher FPR). QwQ-32B HS lead is negative (−1.2 to +1.9), making text detection more timely despite lower accuracy. R1-32B HS achieves the best lead (+11.7 to +13.3 steps) but only at 25% trace coverage. OT-7B lead times are inflated by long traces (∼80 sentences vs. ∼30).

4.9 Oracle Decomposition

Table 7 compares oracle label-based lead times with best probe lead times. The oracle achieves +27 to +31 steps of lead time, while the best probes reach at most +13.3 steps (R1-32B) or +4.5 (R1-8B GRU). Analysis of the R1-8B oracle sequence reveals that labels show gradual erosion (pre-CP label autocorrelation = 0.82, entropy = 1.38), while hidden states exhibit step-function transitions (autocorrelation = 0.95, entropy = 0.41). Hidden states encode current safety state but poorly predict the trajectory toward commitment.

4.10 Cross-Model Transfer

Table 8 shows complete transfer failure. The best cross-model transfer accuracy is 0.673 (R1-32B→QwQ and QwQ→R1-32B), far below within-model performance (0.826 and 0.740 respectively). Cross-architecture family transfers are near chance level, indicating that safety commitment representations are highly model-specific.

4.11 Mahalanobis Distance Baseline

Table 9 shows that Mahalanobis distance fails on all four models, with maximum separation of only 0.20 (QwQ-32B). False alarms exceed true positives at all thresholds. The safety-relevant signal resides in minor PCA components, to which Mahalanobis distance is structurally blind.

4.12 BOCPD and CUSUM

Table 10 shows that BOCPD achieves 0% detection on OT-7B, R1-32B, and QwQ-32B. Only R1-8B shows BOCPD detection (93%), but with shorter lead time (+3.45) than simple first-crossing thresholds (+4.89). CUSUM is universally worse than first-crossing point (FCP) detection. These advanced change-point methods are poorly matched to the signal structure of probe score sequences.

Model	Layer	Safe MD	Post-CP MD	Sep.	FA MD
R1-8B	L14	6.66	6.73	+0.07	7.01
OT-7B	L16	7.68	7.71	+0.03	8.05
R1-32B	L63	6.73	6.88	+0.15	7.07
QwQ-32B	L60	3.22	3.42	+0.20	3.61

Table 9: Mahalanobis distance separation on all models. Maximum separation = 0.20; false alarms exceed true positives at all thresholds.

Model	BOCPD%	BOCPD	CUSUM%	CUSUM
R1-8B	93%	+3.45	65%	+0.89
OT-7B	0%	—	75%	+7.46
R1-32B	0%	—	85%	+2.59
QwQ-32B	0%	—	74%	+9.86

Table 10: Change-point detection results (Det% / Lead). BOCPD fails on 3/4 models (0% detection). CUSUM universally underperforms simple first-crossing (FCP) thresholds.

Model	LR BA	GRU BA	Δ	GRU FPR
R1-32B L63	.737	∼.675	−6.2 pp	.135
QwQ-32B L60	.761	.724	−3.7 pp	.289
OT-7B L16	.688	.746	+5.8 pp	.274
R1-8B L14	.806	.809	+0.3 pp	.165

Table 11: GRU vs. temporal LR at corrected layers. GRU underperforms LR on 2/4 models (32B), confirming linear probes are at the information ceiling.

Model	FCP%	Binary	16-Label	16→Binary	Δ
R1-8B	65%	.817	.41	.825	+0.8 pp
OT-7B	89%	.688	.35	.699	+1.1 pp
QwQ-32B	25%	.740	.26	.749	+0.9 pp
R1-32B	79%	.826	.28	.832	+0.6 pp

Table 12: 16-label vs. binary supervision. Fine-grained labels provide marginal improvement (+0.6–1.1 pp after binary collapse). Transition subtypes are not linearly separable in hidden-state space.

4.13 GRU and Nonlinear Probes

Table 11 shows that GRU sequence models underperform temporal logistic regression on 2/4 models (R1-32B −6.2 pp, QwQ-32B −3.7 pp), likely due to overfitting on small samples. On OT-7B, GRU gains +5.8 pp but at substantially higher FPR (0.274 vs. 0.100). Linear probes appear to be at the information ceiling; additional model complexity does not extract further signal.

4.14 Fine-Grained Taxonomy Supervision

Table 12 shows that 16-label fine-grained taxonomy supervision (Type 1/2/3/4 × pre/post) provides only marginal improvement over binary labels (+0.6 to +1.1 pp after collapsing back to binary). Raw 16-class accuracy is low (26–41%), indicating that fine-grained safety transition types are not linearly separable in hidden-state space.

4.15 False Positive Analysis

Tables 13 and 14 present the false positive analysis. Safe-trace FPR ranges from 11.1% (OT-7B) to 50.0% (QwQ-32B), with a median of approximately 39%. No model simultaneously achieves safe FPR $< 20\%$, precision $> 70\%$, and coverage $> 60\%$, representing a fundamental deployment limitation. On QwQ-32B, HS safe FPR (45–50%) is actually higher than text (42.5%).

5 Discussion

5.1 Summary of Findings

Our results establish three key findings. First, the layer selection metric `bal_acc × crossing_rate` introduces a systematic bias toward shallow layers for 32B models, entirely masking the HS probe advantage that exists at deep layers. Correcting this bias transforms the result from 5/12 to 12/12 Holm–Bonferroni passes. Second, HS probes outperform text probes by +15.4 to +23.6 pp precision at correctly selected layers, but practical deployment is constrained by model-specific timing profiles and high false positive rates. Third, six alternative methods fail comprehensively, establishing supervised temporal linear probing as the current best approach.

5.2 Practical Implications

The precision–coverage trade-off (Table 3) and high safe-trace FPR (Table 13) indicate that current methods are insufficient for production deployment. Cross-model transfer failure (Table 8) means per-model probe training is required, and oracle decomposition (Table 7) reveals that probes detect current safety state rather than predicting future trajectory. These limitations apply to the temporal monitoring pipeline evaluated here and may be exacerbated under distribution shift.

Model	HS Static	HS Temp.	Text Temp.	GRU
R1-8B	.390	.392	.385	.521
OT-7B	.111	.175	.460	—
QwQ-32B	.500	.450	.425	—
R1-32B	.239	.239	.310	—

Table 13: False positive rate on safe traces. Median $\sim$39%. No method achieves >80% precision. Only OT-7B achieves safe FPR < 20%.

Criterion	Threshold	Models	Best
Safe FPR < 20%	0.20	1/4 (OT-7B)	HS static L16
Precision > 70%	0.70	1/4 (R1-8B)	HS temp. W=25
Coverage > 60%	0.60	2/4	HS temp. W=1
All three	—	0/4	None

Table 14: Deployment viability. No model meets all three deployment criteria simultaneously.

5.3 Boundary Condition: AdvBench Non-Replication

To test whether the layer selection bias (C1) is a general property of model geometry or specific to our temporal monitoring pipeline, we replicate the 5-metric layer selection analysis on AdvBench binary classification (R1-8B, $n = 200$ traces: 104 jailbreak, 96 safe).

Tables 15–18 present the AdvBench replication results. The key findings are:

Layer bias does not replicate. On AdvBench, all five metrics select mid-depth layers within a 2-layer range (L12–L14), compared to a 16-layer gap on CP data. The crossing-rate shallow/deep ratio is 1.05 ($p = 0.74$), compared to 2.8 on CP data. This is mechanistically consistent: binary classification lacks the sequential probe-score structure that drives shallow-layer noise inflation.

HS advantage collapses. HS–text precision difference shrinks from $+22.4$ pp (HB pass) on CP data to $+3.0$ pp (NS) on AdvBench. The HS advantage is not a modality-intrinsic property but arises from the interaction between hidden states and temporal step-level monitoring.

These results bound the scope of our methodological contribution: the crossing-rate layer selection bias is specific to temporal monitoring pipelines. Researchers using binary classification or non-temporal evaluation settings will not encounter this issue.

Dimension	CP Temporal	AdvBench Binary
cr composite layer	L0 (0%)	L12 (39%)
threshold metrics layer	L16 (52%)	L13 (42%)
layer gap (abs)	16 layers	2 layers
layer gap (depth %)	52%	6%
CR shallow/deep ratio	2.8	1.05
HS–text Δprec	$+22.4$ pp (HB ✓)	$+3.0$ pp (NS)
HB pass rate	3/3	0/3

Table 15: CP temporal data vs. AdvBench binary: layer selection bias comparison. On AdvBench, all 5 metrics converge to similar layers (gap = 2), crossing-rate ratio is near-uniform (1.05, $p = 0.74$), and HS–text differences are non-significant.

Metric	Layer	Depth	BA at Layer
bal_acc × cr	L12	39%	0.62
pure bal_acc	L13	42%	0.63
precision	L13	42%	0.63
pre_cp_fpr	L14	45%	0.62
composite	L13	42%	0.63

Table 16: AdvBench layer selection (R1-8B, 5 metrics). All metrics select mid-depth layers within a 2-layer range.

5.4 Scope of Contribution C1

The AdvBench non-replication refines C1 from a general observation about hidden-state evaluation to a precise methodological warning: *the layer selection bias arises from the interaction between sequential probe scores and the crossing-rate statistic, and is therefore specific to temporal monitoring settings where threshold-crossing metrics are used for layer selection.* This narrower scope increases precision but reduces the breadth of impact. We view this as a responsible scoping decision: claiming generality that the evidence does not support would be a greater methodological concern than a narrower but well-bounded contribution.

6 Conclusion

We present the first systematic evaluation of detection modalities for reasoning-model safety commitment, spanning four models, two architectures, two scales, and multiple detection paradigms. Our primary methodological contribution—that the `bal_acc × crossing_rate` metric systematically biases layer selection—is specific to temporal monitoring pipelines, as confirmed by AdvBench boundary-condition analysis. At correctly selected layers, HS probes achieve $+15$–24 pp precision over text across all four mod-

Depth Band	Mean CR	Std	n
Shallow (L0–L7)	0.12	0.03	8
Middle (L8–L23)	0.11	0.02	16
Deep (L24–L31)	0.11	0.03	8
Shallow/Deep ratio	1.05	—	—
t-test p	0.74	—	—

Table 17: AdvBench crossing rate by depth band. No significant shallow inflation (ratio = 1.05, $p = 0.74$), unlike CP temporal data where ratio = 2.8.

Modality	BA	Prec	FPR	AUROC
HS (L13)	0.63	0.58	0.31	0.68
Text (MiniLM)	0.60	0.55	0.34	0.65
Δ	$+3.0$ pp	$+3.0$ pp	-3.0 pp	$+3.0$ pp
p-value	0.28	0.31	0.42	0.25
HB pass	No	No	No	No

Table 18: AdvBench HS vs. Text (R1-8B). All differences are $+3$ pp (non-significant, 0/3 HB pass), vs. CP data where Δprec = $+22.4$ pp (3/3 HB pass).

els (12/12 HB), but practical deployment is constrained by high false positive rates ($\sim$39% median on safe traces), model-specific timing tradeoffs, and complete cross-model transfer failure. Six alternative methods fail comprehensively, establishing supervised temporal linear probing with principled layer selection as the current best approach while highlighting fundamental limitations of probe-based safety monitoring.

Limitations

Our evaluation uses commitment-point annotations from a single source, and the temporal monitoring pipeline may not represent all deployment scenarios. The 32B model experiments have limited statistical power ($n = 19$–20), and R1-32B's HS advantage is only significant at $W = 1$ with 25% trace coverage. The AdvBench boundary analysis uses only R1-8B; extending to other models would strengthen the scope-bounding claims.